

Plug Puller

A parametric, 3D-printable tool for pulling a stubborn plug straight out of the wall — shaped to **your** plug and sized to **your** hand. This is the complete how-to: measure, print, and fine-tune the fit.

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Model format: **OpenSCAD (parametric)**

Units: **millimetres (mm)**

License: **PolyForm Noncommercial 1.0.0**

Project & source: github.com/BrennenJohnston/plug-puller-openscad

Everything in this guide is measured in millimetres. If your ruler has two scales, use the mm side — a typical US plug is about 25 mm wide.

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Quick Start for Beginners — Measure, Type Numbers, Print

You do not need to know anything about 3D modeling or code to make a Plug Puller that fits **your** outlet plug and **your** hand. The whole process is:

1. Measure your plug (about 5 minutes) → **2. Type** those numbers into a form and pick a size → **3. Print** the file it gives you.

The default settings reproduce the **reference Plug Puller** flat tool (Medium size, zip ties + wing velcro) — if your plug is a typical two-prong appliance plug, you can even skip the measuring and print the default. In a hurry? Step 1 has a **plug preset** dropdown that fills in the numbers for the three most common US plugs — and for a fat extension-cord plug it switches you to the **heavy-duty clamshell**.

This guide walks you through every click. If a word looks technical, keep reading — the next sentence tells you exactly what to press.

Step A — Get your measurements ready

Open the **Measuring Guide** (it works well on a phone) and write down your numbers on the worksheet at the bottom of that guide. Everything is measured in **millimetres (mm)** — use the mm side of your ruler or caliper.

You will end up with something like this:

What	Example
Plug length (wall plate to the plug's back face)	38 mm
Plug width near the wall	34 mm
Plug width near the cord	34 mm
Plug thickness near the wall	16 mm
Plug thickness near the cord	16 mm
Cord thickness (thin side)	5 mm
Wall plate style	Rocker / Decora
Finger knuckle width (only if measuring your hand)	22 mm
Hand width (only if measuring your hand)	88 mm

(Width and thickness are each measured **twice** — once near the wall, once near the cord — so the tool knows which end of your plug is fatter. On a straight-sided plug like this example the two stations are simply the same number.)

Step B — Install OpenSCAD

OpenSCAD is the free program that turns your numbers into a printable file. You only use one panel of it — you never touch code.

1. Go to <https://openscad.org/downloads.html>.
2. Download the **Development Snapshot** for your system (Windows, Mac, or Linux) and install it like any other program. This project is tested against snapshot `2026.01.03`; any recent snapshot works for personal use. (The regular "stable" release also works — renders are just slower.)

Step C — Open the model

1. Download this project: on the GitHub page, click the green **Code** button → **Download ZIP**, then unzip it anywhere. (Alternative: download just the single file `dist/Plug_Puller_v7_SingleFile.scad` — it is the whole model in one file.)
2. Double-click `Plug_Puller.scad` in the unzipped folder (or open it from inside OpenSCAD with `File ▶ Open`).
3. You may see a wall of code in an editor pane. **Ignore it**. You will not touch it.

Step D — Find the Customizer panel

The Customizer is the form where you type your numbers.

- In the menu bar, open **View** and make sure **Hide Customizer** is **unchecked**. The Customizer panel appears on the right side.
- On older OpenSCAD builds the same panel lives under **Window ▶ Customizer** — tick it there instead.

Step E — Fill in the steps

The form is organized top-to-bottom in the same order you decide things:

0. **Step 0 - Tool Style** — leave `tool_style` on **Auto from plug**. It builds the flat tool for slim plugs and the **heavy-duty clamshell** for fat ones (one serrated plate — print it **twice**, flip one copy over, and zip-tie the pair around the plug). Force **Flat tool** or **Heavy-duty clamshell** only if you want to override.
1. **Step 1 - Your Plug** — pick a **plug preset** (or leave it on **Measure my plug** and type the numbers from your worksheet). Every field's tooltip repeats how to measure it. (These are always active — no mode to switch first.)
2. **Step 2 - Size** — keep **Medium** (the reference Plug Puller size), or pick **Small / Large** for smaller or bigger hands. If you measured your hand, pick **Measure my hand** and type the two hand numbers just below the dropdown.
3. **Step 3 - Attachment** — how the tool straps to the plug: **Zip ties + Velcro** (default), **Zip ties, Velcro strap**, or **None**. Leave `velcro_style` on **Wing** unless you prefer the old rectangular slot.
4. **Step 4 - Cord Hook** — leave `hook_hand` on **Right** (the reference device) or pick **Left** to mirror the cord catch.
5. **You're done — no need to scroll further**. Everything below Step 4 is optional: the `Advanced` - sections are power dials (manual hole placement, clamshell tuning) and the `(Custom size only)` sections only take effect if you set Size to `Custom`. Curious? The [Power User Guide](#) explains them all.
6. The 3D preview updates as you type.

One advanced slider worth knowing: if you're building the heavy-duty clamshell and want a beefier print, `clam_wall_boost` (the first slider in **Advanced - Heavy Duty Clamshell**) thickens every wall of the plate at once — set it to 1–2 mm and everything else adjusts automatically.

How to know it worked: the preview shows a green tag reading `MEDIUM: ...` (or your size) next to the model — that is the model confirming your numbers were applied. If you instead see **red text**, one of your numbers looks wrong (for example, inches instead of millimetres). The red text names which measurement to re-check — fix it before printing. The [Fit Troubleshooting Guide](#) explains every message.

Step F — Render and export the print file

1. Press **F6** (or `Design ▶ Render`). Wait for the progress bar to finish — this takes from a few seconds up to a minute.
2. Then `File ▶ Export ▶ Export as STL...` and save the file.

Call-out: if you export without rendering first, OpenSCAD asks you to render — that's normal. Press F6 and try the export again. The quick preview (F5) *looks* complete but cannot be exported.

Step G — Print it

Load the STL into your slicer with these settings (details in the [README's 3D printing tips](#)):

- **Orientation:** flat face down on the bed, plug pocket facing up. No supports needed.
- **Layer height:** 0.2 mm
- **Walls:** 3–4
- **Infill:** 25–35 %
- **Material:** PETG recommended (PLA works; avoid flexible filament)

Step H — Try it, and fine-tune if needed

Attach the puller to your plug (zip ties or a Velcro strap through the holes), hook the cord in the J-slot, put two fingers in the holes, and pull.

If anything is snug, loose, or uncomfortable, open the [Fit Troubleshooting Guide](#). Every fix is a one-number change: it tells you *which measurement* to nudge and *by how much*, then you repeat steps E–G. Most fits are dialed in within one reprint.

Something not working?

Problem	Fix
I can't find the Customizer panel	Step D — <code>View ▶ Hide Customizer</code> must be unchecked (older builds: <code>Window ▶ Customizer</code>)

Problem	Fix
I type numbers but the model doesn't change	Make sure the <code>size</code> dropdown (Step 2) is not set to <code>Custom</code> — measurements are ignored in Custom mode.
The hand numbers do nothing	They apply only when Size is Measure my hand . The named sizes (Small / Medium / Large) use built-in hand values.
Red text appears in the preview	A measurement is implausible or conflicting. The text names it; see the Fit Troubleshooting Guide .
The exported file is tiny / my slicer says it's empty	You exported after F5 (preview) instead of F6 (render). Press F6, wait, then export again.
A red text tag printed next to my part	That is the warning tag, printed on purpose so a bad file can't fail silently. Read what it says, fix that measurement, re-export, reprint.

Measuring Guide — The Numbers That Make Your Plug Puller Fit

Everything on this page is measured in **millimetres (mm)**. If your ruler or caliper has two scales, **use the mm side** — a US plug is about **25 mm** wide; if you wrote down "1" you measured in inches.

You need:

- a **caliper** (a cheap digital one is ideal) *or* a ruler with mm markings,
- the **plug** you want to pull, plugged into its outlet,
- **your hand** — only if you plan to use the "Measure my hand" size instead of the built-in Small / Medium / Large.

There is also a **printable measuring template** with a mm ruler and finger-sizing circles — print it at 100 % scale ("actual size", no "fit to page") and check the calibration square before trusting it.

Write your numbers into the worksheet at the bottom as you go.

Shortcut: the model's defaults reproduce the reference Plug Puller, which fits a typical 25 mm two-prong plug on a standard flat wall plate. If that describes your plug, you only need to pick a size and can skip the measuring entirely.

Even faster: Step 1 starts with a **plug_preset** dropdown. Pick **Flat 2-prong lamp plug**, **Standard 3-prong plug**, or **Heavy-duty extension cord** and it fills in the plug numbers for you. Leave it on **Measure my plug** to type your own numbers below.

Two tools: Step 0 **tool_style** is **Auto from plug** by default — it builds the flat tool for slim plugs and the **heavy-duty clamshell** (two serrated plates that zip-tie around the plug) for fat ones (plug thickness ≥ 24 mm). So measuring **plug thickness** accurately matters: it decides which tool you get. Force either tool with **tool_style** if you prefer.

Step 1 numbers — your plug and outlet (always used)

If you picked a **plug_preset** other than **Measure my plug**, these six numbers are filled in for you and you can skip to Step 2. Type your own only when the preset is **Measure my plug**.

Most plugs are not the same size at both ends, so the width and the thickness are each measured **twice**: once **near the wall** (just behind the prong face) and once **near the cord** (at the far end of the molded body, skipping any soft rubber cord boot). The two stations tell the tool which end of your plug is fatter, and it shapes the pocket — or the clamshell's gripping arms — to follow that taper in the right direction.

1. Plug length

With the plug in the outlet: ruler from the wall plate face to the plug's back face — how far the whole plug sticks out of the wall. The tool's pocket (or the clamshell's arms) runs this full length.

- Typical range: 20–50 mm
- Example: ≈ 38 mm

2. Plug width near the wall

Caliper straight across the plastic body just behind the prong face, parallel to the wall. Measure the body, not the metal prongs. If the head flares (many plugs have a wider face plate), measure the widest part of that first stretch.

- Typical range: 20–40 mm
- Example: a chunky vacuum plug $\approx$ 34 mm

3. Plug width near the cord

Same direction, but at the far end of the molded body, just before the cord. Skip any soft rubber strain-relief boot — measure the last part of the hard plug body you'd actually grip.

- Typical range: 10–35 mm (most plugs narrow toward the cord)
- Example: $\approx$ 13 mm on a typical lamp plug

4. Plug thickness near the wall

Caliper across the plug body's THIN direction — usually top-to-bottom on a flat plug — **just behind the prong face**. The clamshell grips across this direction, and the bigger of the two thicknesses decides which tool `Auto from plug` builds, so measure carefully.

- Typical range: 12–30 mm
- Example: a flat two-prong plug $\approx$ 16 mm

5. Plug thickness near the cord

Same thin direction, at the cord end of the molded body (again, skip a soft boot). On round extension-cord plugs this can be as fat as the wall end — that is exactly what these two numbers capture.

- Typical range: 8–30 mm
- Example: $\approx$ 9 mm on a lamp plug, 27 mm on a heavy-duty round plug

6. Cord thickness

Caliper across the cord right behind the plug, on its thin side. Flat lamp cord: measure the narrow way. Round cord: the diameter. The cord must slide sideways into the tool's hook slot, so measure where the hook will actually grab — a few centimetres behind the plug.

- Typical range: 3–7 mm
- Example: $\approx$ 5 mm

7. Wall plate style (a picture quiz, not a measurement)

Look at the outlet cover plate and pick the closest match:

Pick this	If your outlet cover looks like
Standard flat plate	The classic plate with two small oval/round openings , one per outlet
Rocker / Decora	One big rectangular opening per outlet (flat "designer" style, like a rocker light switch)
Oversized / Jumbo	Same idea but noticeably bigger and thicker than a normal plate (often used to hide wall damage)

Pick this	If your outlet cover looks like
No plate / flush	No cover plate, or the outlet sits flush with the wall surface

This sets how deep the tool's end notch is, so the tool can straddle the plate and sit flat against the wall.

Where did the side-taper angle go? Earlier versions asked for a `measure_plug_side_angle` in degrees. That angle is now **computed for you** from the two width measurements and the plug length — no protractor required, and the tool automatically knows whether your plug gets wider or narrower toward the cord.

Step 2 — pick a size (hand measurements optional)

The **Size** dropdown covers most hands with built-in values:

Size	Finger width used	Hand width used	Who it fits
Small	16.5 mm	72 mm	smaller adult hands ($\approx$ 5th %ile female)
Medium	20 mm	85 mm	the reference Plug Puller — most adults
Large	23 mm	96 mm	larger hands, gloved use ($\approx$ 95th %ile male)

(The Small / Large values come from ANSUR II 2012 hand-breadth anthropometry, so the built-in sizes span most adult hands.)

Pick **Measure my hand** instead if you want the grip built from your own two numbers:

8. Finger knuckle width

Caliper across the widest knuckle of your middle finger — the finger that goes into the pull hole. Measure the widest point (usually the middle knuckle), not the fingertip.

No caliper? Ring trick: take a ring that fits that finger snugly, measure the ring's **inner** diameter in mm, and add **1.5 mm**.

Or use the template: the [printable template](#) has cut-out circles from $\varnothing$ 15 to $\varnothing$ 32 mm. Find the smallest circle your middle finger passes through comfortably, then **subtract 5** from its $\varnothing$ number — that's your finger width. (Passes $\varnothing$ 26 $\rightarrow$ enter 21.)

- Typical range: 16–24 mm
- Example: $\approx$ 22 mm

9. Hand width

Caliper or ruler across the four knuckles of your flat hand — no thumb. Lay your hand flat, fingers together, and measure straight across the knuckles at the widest point.

- Typical range: 70–100 mm
- Example: $\approx$ 88 mm

Step 3 — pick an attachment (no measuring needed)

How the tool straps onto the plug so it stays put between uses:

Pick this	You get
Zip ties	The four small holes — thread two zip ties around the plug body
Velcro strap	Angled wing slots for a hook-and-loop strap
Zip ties + Velcro (default)	Both sets of openings (the v6 device is hybrid)
None	A clean body (hold the tool on the plug by hand)

Two extra choices sit under Step 3:

- `velcro_style` — `Wing` (default: the v6 curved openings, bigger slot, less plastic) or `Classic slot` (the older rectangular slot).
- `strap_width` — the width of the hook-and-loop strap you'll thread through (10–25 mm; ONE-WRAP comes in 10/13/16/20/25). The wing opening is sized to clear it.

Step 4 — cord hook

Setting	What it does
<code>hook_hand</code> = Right / Left	Which way the J-hook cord catch faces (Right = the reference)

Round / smooth plugs: smooth-sided round cord ends (thick extension-cord plugs) can slip in the pocket. Thread a zip tie down one of the tool's zip-tie holes, around the plug barrel, and back up the opposite hole, then cinch it — the 2×2 hole grid doubles as a clamp anchor.

Safety: the tool touches only the plug's **sides and back** — never between the plug face and the wall. For extra grip on a slick round plug, wrap a turn of silicone tape or a wide rubber band around the ears before clamping.

Worksheet

Print this page (or copy the table) and fill it in. These are the exact names you will see in the form, in the same order.

#	Form field	Your number / choice
—	Plug preset	Measure my plug / Flat 2-prong / Standard 3-prong / Heavy-duty
1	Plug length (mm)	—
2	Plug width near the wall (mm)	—

#	Form field	Your number / choice
3	Plug width near the cord (mm)	—
4	Plug thickness near the wall (mm)	—
5	Plug thickness near the cord (mm)	—
6	Cord thickness (mm)	—
7	Wall plate style	—
—	Size	Small / Medium / Large / Measure my hand
8	Finger knuckle width (mm, only for Measure my hand)	—
9	Hand width (mm, only for Measure my hand)	—
—	Attachment	Zip ties / Velcro strap / Zip ties + Velcro / None
—	Velcro style / strap width	Wing / Classic slot · ____ mm
—	Cord hook hand	Right / Left

Sanity check before you continue: each plug width should be a two-digit number (roughly 12–45), and the wall-end width is usually the bigger one. Finger width should be roughly 14–32. If a number looks like `1.3`, it's probably inches — re-measure with the mm side.

Next step: the **Quick Start**, Step B.

Example filled-in worksheets

Example measured devices ship as saved Customizer parameter sets in

`v7.0/presets/Plug_Puller_v7_Parametric.json` (in OpenSCAD: Customizer panel → the preset-set dropdown above the sections → pick a set):

Set name	The values
Medium (v6 reference)	all defaults — the v6 device, zip ties + wing velcro
Flat 2-prong lamp plug (NEMA 1-15)	the lamp preset (37 mm long, widths 25 → 11.2, thickness 18.6 → 8.6, cord 3.6)
Standard 3-prong plug (NEMA 5-15)	the standard preset (46.2 mm long, widths 26.6 → 13.4, thickness 18.9 → 15, cord 7)
Heavy-duty round cord (NEMA 5-15)	the heavy-duty preset (43.8 mm long, 27 mm thick at both ends → clamshell, cord 8.2)
Measure my plug + hand (US vacuum plug)	straight-sided plug 34 wide / 16 thick at both stations, 38 mm long, cord 5 mm, Rocker / Decora, finger 22 mm, hand 88 mm
Left-handed + classic velcro slots	<code>hook_hand = Left</code> , <code>velcro_style = Classic slot</code>

Load one to see what a completed form looks like, then overwrite the values with your own numbers.

When you measure for someone else

If the person who will *use* the tool is not the person *measuring* (a relative, a client of an occupational therapist), measure **their** hand for numbers 8–9 and **their** outlet and plug for numbers 1–7. Doubtful between two values? Round **up** — a slightly roomy fit still works; a tight one doesn't.

Fit Troubleshooting — One Measurement, One Nudge, One Reprint

Printed your Plug Puller and something isn't quite right? Find your symptom below. Every fix names **one measurement** to change and **by how much**. Change it in the Customizer form, re-export (F6, then **File ▶ Export ▶ Export as STL...**), and reprint. You never need to touch anything below the three Step groups at the top of the form.

Rule of thumb: nudge in small steps (0.5–2 mm). Looser always beats tighter — a slightly roomy tool still works, a tight one doesn't.

Physical fit problems

#	Symptom	Change this	By how much
1	Plug body rubs the sides of the end notch / won't slide in	Plug width near the wall	+1 mm
2	Plug falls out of the notch or rattles side to side	Plug width near the wall	−0.5 mm
3	Plug won't seat down into the pocket	Plug length	+2 mm
4	Plug seats, but the tool overhangs the plug by a lot (pocket much longer than the plug)	Plug length	−2 mm
5	Plug sits proud of the pocket / tool rocks on top of the plug	Plug thickness (the station that measured fatter)	+2 mm
5b	Pocket walls pinch the plug's cord end (or gape at it)	Plug width near the cord	+1 mm (pinches) / −1 mm (gapes)
6	Hook won't take the cord / cord has to be forced in	Cord thickness	+0.5 mm
7	Cord slips out of the hook while pulling	Cord thickness	−0.5 mm
8	Tool won't sit flat against the wall — the outlet cover pushes it away	Wall plate style	Pick the next bigger style (Standard flat plate → Rocker / Decora → Oversized / Jumbo)
9	Fingers pinch in the holes / knuckle drags on the rim	Size up one step — or switch to Measure my hand and add +1 mm finger knuckle width	
10	Fingers swim in the holes and the pull feels sloppy	Size down one step — or Measure my hand with −1 mm finger width	

#	Symptom	Change this	By how much
11	Tool edges dig into your palm / tool feels too wide	Size down — or Measure my hand with –5 mm hand width	
12	Tool feels too small in the hand / fingers crowd the edges	Size up — or Measure my hand with +5 mm hand width	
13	Tool too wide for a cramped outlet corner (hits trim or another plug)	Size down one step (the body tracks the hand size)	
14	Strap keeps sliding off the plug	Attachment → <code>Zip ties + Velcro</code> and use both	
15	Smooth round plug still slips out of the zip-tie hold	Thread a zip tie down one zip-tie hole, around the plug barrel, and back up the opposite hole, then cinch it — the 2×2 hole grid doubles as a clamp anchor	
16	Hook is on the wrong side for your hand	Step 4 → <code>hook_hand</code> = Left (or Right)	
17	Strap won't thread through the wing opening	Step 3 → lower <code>strap_width</code> , or use a narrower strap (the wing is sized to it)	

Red warning tags

If the preview shows red text — or a red text tag **printed next to your part** — the model detected a problem before you wasted a full print. That's deliberate: a bad file fails loudly instead of silently. Read the tag, fix that measurement, re-export.

Tag says	What it means	What to do
<code>CHECK <NAME> MEASUREMENT (MM?)</code>	That number is outside any plausible mm value — classic inch entry (e.g. 1.25 instead of 32)	Re-measure with the mm side, re-type
<code>FINGER TOO BIG FOR HAND WIDTH - RECHECK BOTH</code>	The finger holes can't physically fit inside a body sized for that hand width	Re-measure both finger knuckle width and hand width — one of them is off
<code>PLUG TOO WIDE FOR THIS DESIGN (MAX 38MM)</code>	Your plug (at the wall end) is wider than the tool's end can open up	The design tops out near 38 mm plug width; if your plug is really that wide, this tool geometry can't grip it
<code>PLUG LONGER THAN POCKET LIMIT - POCKET SHORTENED</code>	The plug is longer than the pocket budget inside the tool's 120 mm body ceiling, so the pocket was truncated — the tool still works but won't swallow the whole plug	Double-check plug length; if it's real, print and try it
<code>PLUG WIDTH TAPER TOO STEEP - RECHECK BOTH WIDTHS</code>	The two width measurements describe a taper steeper than the pocket walls can follow	Re-measure plug width near the wall and near the cord — one of them is probably off
<code>CORD TOO THICK FOR HOOK SLOT</code>	The hook slot can't open wide enough for that cord	Re-measure the cord's thin side; cords past ~9 mm don't fit the hook

Tag says	What it means	What to do
FINGER HOLES HIT PLUG POCKET	Your finger + plug combination makes the holes collide with the plug pocket	Reduce finger knuckle width slightly, or re-check plug length
FINGER HOLES TOO CLOSE - WEAK BRIDGE	The strip of plastic between the two holes is too thin to be strong	Reduce finger knuckle width by 1 mm
FINGER HOLES OUTSIDE BODY - INCREASE HAND WIDTH	The holes for your fingers don't fit inside a body sized for your hand width	Increase hand width (or re-measure both)
PLUG TOO WIDE FOR TOOL END - CHECK PLUG WIDTH	The end notch came out wider than the tool's end itself	Re-measure plug width — it's probably too large
PLUG SEAT OVERHANGS BODY - RECHECK PLUG WIDTH / PLUG POCKET OVERHANGS BODY ...	The plug pocket is wider than the tool body at that spot	Re-measure plug width, or go up a size so the body grows
WING OPENING SMALLER THAN STRAP WIDTH	The wing velcro slot is too narrow to pass the strap you set	Lower <code>strap_width</code> (Step 3), or switch <code>velcro_style</code> to <code>Classic slot</code>
WING WEB COLLAPSED - NO ROOM FOR STRAP	Features crowded the wing until no opening is left	Go up a size (bigger body) or lower <code>strap_width</code>
ZIP TIE HOLES HIT FINGER HOLES	A zip row landed too close to a finger hole (with <code>Auto</code> placement the rows now derive around the finger holes, so this fires for <code>Manual</code> <code>zip_pos_*</code> dials or Custom mode with auto-fit off)	Move the offending <code>zip_pos_*</code> dial, re-enable auto-fit, or lengthen the body
ZIP TIE ROWS OVERLAP EACH OTHER	Two manual zip rows landed on top of each other	Spread the <code>zip_pos_*</code> dials at least one hole diameter apart
ZIP TIE HOLES HIT VELCRO SLOTS	A zip row broke into a classic velcro slot	Move <code>zip_pos_*</code> or <code>velcro_pos</code> apart (Step 3)
SEAT HAS NO RECESS ... / POCKET FLOOR TOO THIN ... / anything else	An internal geometry check — with measured inputs this shouldn't happen	Re-check all numbers against the Measuring Guide ; if it persists, open an issue with your numbers
A green <code>MEDIUM: ...</code> / <code>MEASURED: ...</code> tag in the preview	Not a warning! Positive confirmation that your size and numbers were applied	Nothing — it never appears in the exported file

Heavy-duty clamshell tags

These appear only when you're building the clamshell (a thick plug, or `tool_style = Heavy-duty clamshell`):

Tag says	What it means	What to do
CORD TOO THICK FOR CABLE CHANNEL	The cord won't fit the channel with clearance	Re-measure the cord thickness, or raise <code>clam_cable_clearance</code>
PLUG TOO THICK - ARMS BULGE PAST FINGER LOBES	The plug is so thick the tapered arms would bulge wider than the finger lobes	Re-check plug thickness; a truly huge plug may exceed this tool

Tag says	What it means	What to do
NO GRIP BITE - PLUG WONT BE HELD	<code>clam_grip_bite</code> is ≥ 0 , so the arms don't squeeze	Set <code>clam_grip_bite</code> negative (e.g. -1) so the arms bite the plug
PLATE THINNER THAN 2MM - TOO FLIMSY	<code>clam_plate_thickness</code> is under 2 mm	Raise <code>clam_plate_thickness</code> to 3–5 mm
ZIP STATION OFF THE ARM	A manual clamshell zip position fell past the arm	Bring <code>clam_zip_pos_*</code> back within the arm length
ZIP STATIONS OVERLAP EACH OTHER	Two manual clamshell zip positions landed on top of each other	Spread <code>clam_zip_pos_*</code> at least one hole diameter apart
ZIP STATION HITS VELCRO SLOT	A manual clamshell zip position broke into the velcro slot	Move <code>clam_zip_pos_2/3</code> — in <code>Auto</code> placement the slot always keeps clear of the stations
PLUG TOO LONG - PLATE OVER 120MM, CHECK PLUG LENGTH	The arms grow with the plug, and this plug length pushed the plate past 120 mm	Re-measure plug length (wall plate to the plug's back face — not including the cord)
PLUG THICKNESS TAPER LOOKS WRONG - RECHECK BOTH ENDS	The two thickness measurements are more than $\sim 20^\circ$ of taper apart — almost certainly a mis-measurement	Re-measure plug thickness near the wall and near the cord

Clamshell feels too flimsy? Raise `clam_wall_boost` (the first slider in the **Advanced - Heavy Duty Clamshell** section): it thickens **every** wall around the inner openings at once — finger-hole walls, the cord-channel web, both velcro-slot walls, and the zip-tie webs — while the automatic placement keeps all the holes clear of each other. For just the wall between the gripper teeth and the velcro slot, raise `clam_slot_inner_wall`. More clamshell dials: the [Power User Guide](#).

Printing problems (not fit problems)

Symptom	Cause	Fix
Slicer complains about floating/disconnected parts	You exported with a red warning tag present — the tag is a separate object	Fix the named measurement first; a clean export has exactly one body
Layers split at the pocket floor	Under-extrusion or too few walls	3–4 walls, 25 %+ infill; see the README printing tips
Tool snapped at the hook while pulling	Printed in a brittle material or too-thin walls	Reprint in PETG with 4 walls (clamshell: raise <code>clam_wall_boost</code> and/or <code>clam_plate_thickness</code>)

Still stuck?

Re-do the 5-minute [Measuring Guide](#) from scratch — most stubborn misfits trace back to one mis-read number. If the fit still fails after two nudges, open a GitHub issue with your inputs and a photo of the tool on the plug; that's exactly the feedback that improves the fit formulas for everyone.

Power User Guide — The Advanced and Custom Tiers

This guide is for experienced makers, occupational therapists tuning fits for clients, and engineers who want every dial the model offers. It assumes you already know the basic flow (open the file, fill in Steps 0–4, F6, export) — if not, start with the [Quick Start for Beginners](#).

The Customizer is organized in **three tiers**, top to bottom:

Tier	Sections	Who it's for
Guided steps	Step 0 ... Step 4	Everyone. Measurements + four dropdowns; the model derives all geometry from these.
Advanced	Advanced - Zip Tie Placement - Flat Tool, Advanced - Velcro Placement - Flat Tool, Advanced - Heavy Duty Clamshell, Advanced - Render Quality	Power users. Placement overrides and tuning dials that work <i>with</i> the measured sizes — no mode switch needed.
Custom	Custom Mode + every section marked (Custom size only)	Experts. Set Step 2's size = Custom and the measurements are ignored entirely; ~50 sliders control the raw geometry.

Everything in the Advanced tier applies immediately, in any size. The Custom tier is inert until size = Custom — if you move one of those sliders in a measured size, the preview shows an orange CUSTOM SLIDERS IGNORED tag and the console names each ignored slider.

1. Workflow tools

Saved parameter sets

The Customizer panel has a preset bar above the sections: + saves your current values as a named set inside a JSON next to the .scad file. Ship-with examples live in v7.0/presets/Plug_Puller_v7_Parametric.json. Sets are plain JSON — versionable, diffable, and shareable.

Command-line batch export

Every parameter can be overridden from the CLI, which makes A/B testing and batch generation scriptable:

```
# Render the heavy-duty clamshell plate with a 2 mm strength boost
openscad -o plate.stl --backend Manifold \
  -D 'plug_preset="Heavy-duty extension cord - NEMA 5-15"' \
  -D 'render_mode="Clamshell Plate"' \
  -D clam_wall_boost=2 \
  v7.0/src/Plug_Puller_v7_Parametric.scad

# Apply a saved parameter set
openscad -o out.stl -p v7.0/presets/Plug_Puller_v7_Parametric.json \
  -P "Left-handed + classic velcro slots" \
  v7.0/src/Plug_Puller_v7_Parametric.scad
```

The hidden `render_mode` switch

`render_mode` lives under `/* [Hidden] */` (set it via `-D` or by editing the declaration) and renders geometry subsets — useful for debugging a single feature or documenting a change:

Value	Renders
<code>Full</code> (default)	The resolved tool + warning tags
<code>Clamshell Plate</code>	The heavy-duty plate regardless of <code>tool_style</code>
<code>Body Only</code> / <code>Body No Cutouts</code>	Flat-tool body variants
<code>Only Finger Holes</code> / <code>Only T Hook</code> / <code>Only Plug Wall Notch</code> / <code>Only Zip Tie Holes</code> / <code>Only Velcro Strap Holes</code>	Flat body + one cutout
<code>Cutouts Only 2D</code>	A flat 2D overlay of every cutout + pocket profile

Console diagnostics

Every render echoes a **Final Adapted Dimensions** block: each derived value, with `(clamped from <your value>)` appended wherever auto-fit had to adjust an input. Non-Custom sizes also echo one `fit_derived: <key>=<value>` line per derivation, which is what the CI parity tests consume.

2. Flat tool power dials

The plug side rail

One concept powers the flat tool's Advanced tier: a 2D **rail** that runs down the plug's side, starting at the pocket edge on the plug face and sloping by the plug's side-taper angle. The angle is **derived from the two Step 1 width stations** over the plug length (in Custom mode `custom_pocket_side_angle` sets it directly), so its sign encodes the taper direction: a plug that is wider at the cable end gives a negative angle and the pocket walls widen toward the cord. The pocket walls, the zip-tie stations, and the manual velcro slots all hang off this rail, so their positions are measured in **mm along the plug's side, from the plug face toward the cord** — the taper moves everything coherently.

Zip-tie placement (Advanced - Zip Tie Placement - Flat Tool)

- `zip_placement = Auto` spaces `zip_row_count` (1–3) hole pairs along the rail and automatically compresses the spacing rather than letting a row punch into the finger holes.
- `Manual` places each pair with `zip_pos_1/2/3` (mm along the rail). Manual placements are *not* collision-protected — the red warning tags (`ZIP TIE HOLES HIT FINGER HOLES`, `...OVERLAP EACH OTHER`, `...HIT VELCRO SLOTS`) fire if you overlap something.
- `zip_edge_offset` slides the whole column toward/away from the pocket wall.

Velcro placement (Advanced - Velcro Placement - Flat Tool)

`velcro_placement = Manual` slides a pair of classic slots along the rail with `velcro_pos`. The `Wing` style ignores manual placement (its opening is derived from the body's dead space); switch `velcro_style = Classic slot` in

Step 3 to use it.

3. Heavy-duty clamshell tuning (`Advanced - Heavy Duty Clamshell`)

The clamshell derives its core dimensions from the Step 1 measurements: the arm's gripping edge follows the plug's own **two-station thickness profile** (thickness near the wall at the arm tips, thickness near the cord at the plug's back end, interpolated in between), the arm length grows so the arms always cover the full plug length, the throat — where the V closes down to the cord channel — sits just below the plug's back end, the cord channel comes from the cord thickness, and the finger bores from the Step 2 hand. The `clam_*` dials tune that result. The defaults carry the field-tested **T3 grip profile** (2 mm teeth, 2.8 mm pitch, 1 mm bite, grip bite -1) with the serration span on auto.

Strength recipes

Goal	Dial(s)
Denser, stronger plate all around	<code>clam_wall_boost</code> — adds N mm to every wall around the inner openings at once (finger walls, cord-channel web, both velcro-slot walls, zip webs). The outline grows and the holes re-place themselves; Auto placement keeps every web intact at any boost.
Stiffer sandwich	<code>clam_plate_thickness</code> (each half; the stack is 2×)
Beef up only the tooth-to-slot boundary	<code>clam_slot_inner_wall</code> — measured from the deepest tooth bite , so bigger teeth never silently thin it
Less plastic / lighter	<code>clam_velcro_slot_width/length</code> (the slots double as material reduction), <code>clam_strip_thickness</code> = 0

Grip tuning

Goal	Dial(s)
Squeeze harder / looser	<code>clam_grip_bite</code> (negative = interference squeeze; applied per side on top of the plug's own thickness profile)
Teeth that bite soft plug bodies	<code>clam_tooth_diameter</code> , <code>clam_tooth_pitch</code> , <code>clam_tooth_depth</code>
Where the teeth sit	<code>clam_grip_zone_start</code> / <code>clam_grip_zone_length</code> (measured back from the arm tips; length 0 = auto — the teeth cover the full plug body span)
Easier plug entry	<code>clam_tip_flare</code> (opens the V at the tips)
Cord slides freely	<code>clam_cable_clearance</code>
Finger security	<code>clam_finger_fit</code> (bore = finger width + this)

Manual zip placement (`clam_zip_placement` = Manual + `clam_zip_pos_1/2/3`) trades the collision guarantees for full control; the `wc-*` warning tags call out stations that fall off the arm, overlap each other, or break into the velcro slot.

4. Custom mode — the full unlock

Set **Step 2** `size = Custom` and the measurement/derivation layer switches off: every slider in the `(Custom size only)` sections drives the flat-tool geometry directly (body octagon, pocket, finger holes, J-hook, notch, zip grid, velcro, edge rounding).

- `reset_custom_to_medium` — render once with this on to snap the output back to the Medium reference geometry (a known-good baseline), then turn it off and diverge.
- `custom_enable_auto_fit` — on (default), every slider is clamped so features stay inside the body with printable webs; each clamp is echoed as `(clamped from ...)` and the preview shows an `AUTO-FIT ADJUSTED N VALUES` notice. Off = raw values, no protection; the warning tags become your only guard rail.
- The Step 3/4 dropdowns (attachment, velcro style, strap width, hook hand) remain authoritative even in Custom.

The full slider matrix — names, types, ranges, steps, defaults — is the machine-readable schema `v7.0/parameter_mapping.json` (103 parameters, CI-validated against the SCAD).

5. Guard rails and how to read them

The model never silently "fixes" your input past what auto-fit reports. Anything genuinely wrong prints a **red warning tag** flat on the bed next to the part (so an exported file physically shows its own defect) and echoes the same message to the console. The [Fit Troubleshooting guide](#) decodes every tag — flat-tool tags `W-1...W-19`, clamshell tags `WC-1...WC-9`.

Preview-only (never exported): the green `MEDIUM: ... / MEASURED: ...` confirmation tag, and the orange HUD notices for ignored custom sliders and auto-fit clamps.

6. Going deeper

Resource	What's in it
<code>v7.0/docs/Plug_Puller_v7_Reference.md</code>	The engineering reference: coordinate frames, the rail math, clamshell derived values, CSG order, derivation layer, parity invariants
<code>v7.0/parameter_mapping.json</code>	Machine-readable Customizer schema (source of truth for the validator)
<code>docs/RENDERING_POLICY.md</code>	<code>\$fn</code> / <code>quality</code> tessellation policy
<code>tests/</code>	The pytest + OpenSCAD framework: golden-fixture mesh parity, clamshell parity vs the idealized plate, derivation-formula mirror
<code>scripts/</code>	<code>regenerate_fixtures.py</code> , <code>build_flattened.py</code> , <code>generate_parameter_mapping.py</code> — keep these in sync when you change the SCAD

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Appendix — Printable Measuring Template

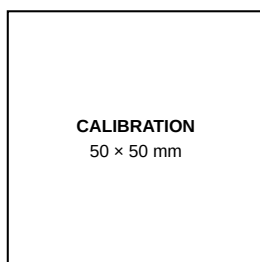
The measuring template is a one-page sheet with a millimetre ruler, a stop bar for reading plug widths, finger-sizing circles, and a **50 mm calibration square** for confirming your printer output is true size. The [Measuring Guide](#) explains how to use it.

Print the companion file — not this page. For a guaranteed 1:1 scale, print

`Plug_Puller_Measuring_Template.pdf` (the separate one-page file that accompanies this guide) at **100%** / **"Actual size"** — never "Fit to page" — on A4 or US Letter paper. Then measure the calibration square: it must read exactly **50 × 50 mm** before you trust any measurement taken from the sheet. The preview below is scaled down to fit this page and is **not** to scale.

Plug Puller — Measuring Template

Print at 100% scale / "Actual size" — never "fit to page". Check the calibration square below before trusting anything on this sheet.

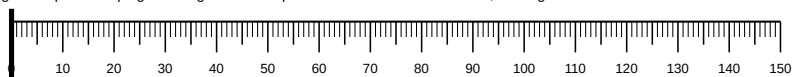


1. Measure this square with a ruler.
It must be exactly 50 × 50 mm.
If it is not, your print was scaled —
re-print at 100% before using this sheet.

Works on A4 and US Letter paper.
Full instructions: <docs/guides/measuring-guide.md>

2. Ruler (mm) — plug length, plug widths, cord, hand width.

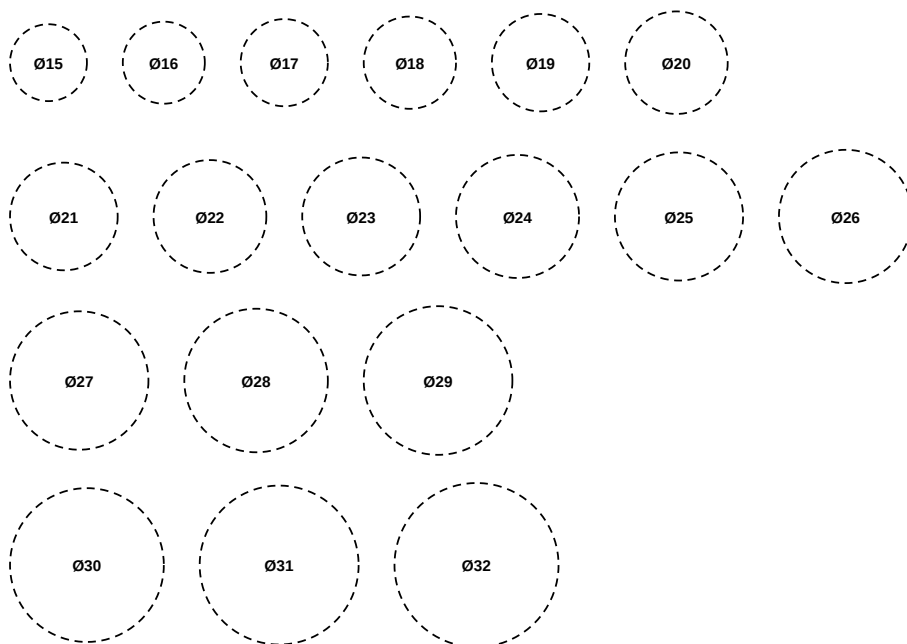
Plug width: press the plug's side against the stop bar — read it near the wall end, then again near the cord end.



3. Finger-sizing circles — cut out along the dashed lines.

Find the smallest circle your middle finger passes through comfortably.

Your finger width = that circle's Ø number minus 5. (Example: passes Ø26 → enter 21.)



plug-puller-openscad - measuring template - guide: <docs/guides/measuring-guide.md> - troubleshooting: <docs/guides/fit-troubleshooting.md>

Preview only — not to scale. Print the companion PDF [Plug_Puller_Measuring_Template.pdf](#) at 100% and verify the 50 mm calibration square.